

Seasonal Dynamism or Realistic Models: What Drives Better Predictions of Landscape Connectivity?

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Abstract

Connectivity modeling has become a cornerstone of conservation science. Most studies investigating functional landscape connectivity assume static spatial covariates and fixed species-specific habitat preferences. That is, they assume the landscape remains unchanged over time and that species exhibit stable preferences with respect to their surroundings, thereby overlooking seasonal dynamics. Additionally, connectivity studies typically rely on simplistic species-habitat associations rather than realistic movement models that mechanistically incorporate movement. These limitations may mask key ecological processes, leading to poor agreement between predicted and observed movements and a misrepresentation of connectivity.

We develop a conceptual framework illustrating that seasonality can be incorporated at three stages of the connectivity modeling pipeline: (1) when extracting spatial covariates for model fitting, (2) when specifying the selection model, and (3) when making predictions from the fitted model, yielding several configurations that differ in their degree of seasonal dynamism. Further, we compare the performance of alternative models that vary in their degree of biological realism, ranging from simplistic habitat-focused models to realistic formulations that incorporate not only interactions with the physical environment but also non-geographic factors likely to influence movement behavior.

Capitalizing on natural seasonal fluctuations of the Okavango Delta in northern Botswana and on GPS data collected on dispersing individuals, we assess to which extent incorporating seasonality improves our ability to predict dispersal trajectories and connectivity for the endangered African wild dog *Lycaon pictus*. For this, we fit integrated step-selection functions and predict connectivity using individual-based dispersal simulations while explicitly considering seasonal dynamism in both environmental covariates and the species' habitat and movement preferences. At the same time, we compare predictions from a simplistic model with predictions from a mechanistic and biologically realistic model, enabled through individual-based simulations from step-selection functions.

Through cross-validation, we evaluate predictive performance across various seasonal configurations, and between simplistic and realistic movement models. We hypothesized that incorporating seasonality and adopting a more realistic view would improve predictions.

Contrary to expectations, our results showed that, while accounting for seasonality across all three stages of the modeling pipeline moderately improved predictions in the simplistic model, its benefits disappeared in the realistic approach, which already achieved strong predictive performance. Our study provides a conceptual framework for integrating seasonality into connectivity analyses and shows that, for our study

system, a more realistic representation of movement outperformed the incorporation of seasonality. This suggests that connectivity studies should prioritize realistic movement models over simplistic approaches while carefully considering when and how to account for seasonal dynamism.

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1 Introduction

1.1 Connectivity

Landscape connectivity is defined as the degree to which the landscape facilitates or impedes movement among habitat patches (Taylor et al., 1993) and is a central prerequisite for maintaining biodiversity (Fahrig, 2003). Improved connectivity facilitates dispersal (Dorr et al., 2011; Baguette et al., 2013), which in turn promotes genetic diversity (Perrin and Mazalov, 2000; Frankham et al., 2002) and the colonization of vacant habitats (Hanski, 1999; MacArthur and Wilson, 2001). Due to its beneficial impacts on metapopulation dynamics, restoring connectivity is among the most frequently recommended strategies to conserve biodiversity and to promote resilience against climate change (Heller and Zavaleta, 2009; Rudnick et al., 2012). Quantifying connectivity and identifying critical dispersal corridors have therefore become crucial tasks in conservation science (Heller and Zavaleta, 2009; Rudnick et al., 2012; Keeley et al., 2019; Hofmann et al., 2021). Reliable estimation of connectivity is thus paramount for a targeted allocation of scarce conservation funds. Two factors that can hinder our ability to predict connectivity are disregarding seasonality and relying on overly simplistic models that fail to capture more mechanistic aspects of movement behavior (Osipova et al., 2019; Zeller et al., 2020a; Hofmann et al., 2023).

1.2 Seasonality

Seasonality can impact functional connectivity through spatio-temporal variation in the landscape itself, or through temporal variation in species' preferences towards prevailing conditions (Mui et al., 2017; Simpkins and Perry, 2017; Zeller et al., 2020a; Forrest et al., 2024). In ecosystems that experience alternations between wet and dry seasons, for example, the onset of the rainy season initiates distinct “green-up” waves, which affect the availability of food resources for herbivores and, subsequently, shape their movements (Merkle et al., 2016). In its most remarkable form, the variation in environmental conditions drives herbivore migrations across massive spatial scales, resulting in short-lived movement corridors between otherwise disconnected habitats (Serneels and Lambin, 2001; Naidoo et al., 2016). Alternatively, seasonality can affect functional connectivity via temporal changes in a species' habitat preferences. Amphibians, for instance, require both aquatic and terrestrial habitats, but their preference for one habitat over the other heavily depends on the season (Baldwin et al., 2006). Although such seasonal intricacies are likely to play a fundamental role in many ecosystems, they only rarely enter connectivity studies in an explicit manner. In fact,

most connectivity studies represent their study system by a static set of environmental layers and assume their focal species to exhibit a fixed set of preferences (e.g., Elliot et al., 2014; Abrahms et al., 2017; Brennan et al., 2020, but see Forrest et al., 2024). Overcoming these limitations requires viewing connectivity as a dynamic entity and modeling it accordingly (Zeller et al., 2020a).

1.3 Modeling Functional Connectivity

A second challenge in accurately predicting connectivity arises from the use of overly simplistic models that fail to capture the complexity of animal movement. Traditional approaches to study functional connectivity generally follow a three-step pipeline: (1) compiling presence or movement data for the focal species, preferably during dispersal (Elliot et al., 2014; Vasudev et al., 2015; Benz et al., 2016, but see Fattebert et al., 2015), alongside spatial covariate layers that influence connectivity, (2) fitting a selection model to estimate species preferences for environmental features using techniques like point-selection functions, path-selection functions or step-selection functions (Boyce et al., 2002; Manly et al., 2007; Cushman and Lewis, 2010), and (3) generating a resistance (or permeability) surface based on inferred preferences, which is then input into a connectivity model (Zeller et al., 2012; Diniz et al., 2019). One downside to such resistance-based approaches is that associated models are limited to effects that can be represented as spatial layers. This prevents scientists from including biologically relevant, yet non-spatial effects. More recently, a growing alternative has been to bypass the resistance surface and simulate individual movement directly from the selection model, thus inferring connectivity from emergent movement patterns (Kanagaraj et al., 2013; Allen et al., 2016; Hauenstein et al., 2019; Zeller et al., 2020b; Unnithan Kumar et al., 2022b,a; Hofmann et al., 2023). This shift has been accelerated by the rise of integrated step-selection functions (iSSFs), which allow jointly modeling an animal’s movement kernel and habitat-selection function (Avgar et al., 2016; Fieberg et al., 2021) and allow simulating movement directly from a fitted model (Avgar et al., 2016; Signer et al., 2017). Unlike resistance-based approaches, iSSF-based simulations are not limited to effects that can be represented raster cells and offer greater mechanistic realism by capturing non-spatial, dynamic factors—such as directional persistence, circadian rhythms, and interactions between movement behavior and environmental variables (Hofmann et al., 2023; Potts and Börger, 2023; Signer et al., 2024; Forrest et al., 2024). Furthermore, mechanistic simulations open up new avenues for incorporating seasonality as they eliminate the need for seasonally updated resistance surfaces and allow incorporating seasonality directly in the

simulation.

1.4 Incorporating Seasonality in Functional Connectivity Models

Accounting for seasonality in connectivity analyses is non-trivial, as seasonality can enter at three distinct stages of the connectivity modeling pipeline (Figure 1). (1) When compiling spatial covariates (e.g., vegetation, water cover), researchers can either use a single layer per covariate—representing a static or averaged snapshot—or a series of layers that capture seasonal variation (Figure 1, Stage 1). Historically, static representations have been the norm (e.g., Elliot et al., 2014; Brennan et al., 2020). However, advances in remote sensing and increased access to petabytes of landscape data (Toth and Józków, 2016; Rumiano et al., 2020) have made high-resolution spatiotemporal layers more accessible, enabling more dynamic representations of study systems (e.g., Osipova et al., 2019; Kaszta et al., 2021; Forrest et al., 2024). (2) When fitting a selection model, species’ habitat preferences can either be assumed constant by pooling all data—ignoring seasonal variation—or modeled separately by season to account for changing preferences (Figure 1, Stage 2; e.g., Fortin et al., 2005; Manly et al., 2007; Cushman and Lewis, 2010; Zeller et al., 2020b). For instance, Chetkiewicz and Boyce (2009) partitioned their data by season to estimate seasonal habitat preferences for pumas (*Puma concolor*) and grizzly bears (*Ursus arctos*). (3) Finally, one must decide whether to predict connectivity for “average” environmental conditions or for distinct seasons (Figure 1, Stage 3). Using resistance-based models, the latter approach requires repeatedly applying connectivity models to seasonally updated resistance surfaces (e.g. Osipova et al., 2019; Zeller et al., 2020b; Kaszta et al., 2021; Ciudad et al., 2021). This is demonstrated *ad extremum* by Kaszta et al. (2021), who prepared monthly updated resistance surfaces and employed circuit theory to estimate dynamic connectivity for African elephants (*Loxodonta africana*). iSSF-based simulations offer a more seamless alternative, as seasonality can enter the simulations directly and the environment or species’ preferences can be adjusted as simulated individuals move. However, such an approach has not yet been employed. Regardless of the method, integrating seasonality into connectivity analyses remains analytically and computationally demanding (Bishop-Taylor et al., 2018), raising the question of when and to what extent it should be prioritized.

1.5 African wild dogs in Northern Botswana

To study the importance of seasonal dynamism and a mechanistic representation of movement when predicting connectivity, we use a population of African wild dogs (*Lycaon pictus*)

inhabiting the highly seasonal Okavango Delta ecosystem in northern Botswana as a study system (McNutt, 1996). While once present across the entire Sub-Saharan continent, the African wild dog has disappeared from a majority of its historic range due to human persecution, deadly diseases, and habitat destruction (Woodroffe and Sillero-Zubiri, 2020). With about 6,000 adult individuals remaining in the wild, the species is considered as endangered on the IUCN red list. Wild dogs are pack-living carnivores, primarily active during the cooler morning and evening hours (Rasmussen and Macdonald, 2012) and during moonlit nights (Cozzi et al., 2012). Higher ambient temperature is associated with shorter activity periods, as the species usually rests during times of elevated heat (Rabaiotti and Woodroffe, 2019). Upon reaching sexual maturity, individuals born into a pack disperse in same-sex coalitions to find suitable mates and a territory to settle (McNutt, 1996). In Botswana, the timing of dispersal is seasonal, with female dispersal peaking prior to the mating season in March, and male dispersal peaking at the onset of the rainy season in December (Behr et al., 2020). Euclidean distances moved by dispersers range from a few dozen to 500 km, with some coalitions covering several hundred kilometers within only a few days (Davies–Mostert et al., 2012; Masenga et al., 2016; Cozzi et al., 2020; Sandoval-Seres et al., 2022). Studies of habitat-selection show that dispersing wild dogs avoid water, prefer moving along it, prefer moving across open grassland or shrubs, yet avoid areas dominated by humans and densely covered by forests (O’Neill et al., 2020; Hofmann et al., 2021). Given the species’ considerable mobility and the study area’s significant seasonal variability, we deemed the study-system well-suited to examine the importance of seasonal dynamism and mechanistic models when examining connectivity.

1.6 Study Goals

Our aims are to (1) introduce a framework for incorporating seasonality into connectivity analyses, (2) evaluate whether including seasonality improves the predictive performance of dispersal models, and (3) assess the value of more mechanistic representations of movement when predicting connectivity. To achieve this, we compare the predictive performance of six seasonal configurations (Figure 1b) and of a simplistic and mechanistic movement model. Specifically, we examine how incorporating seasonality at various stages of the connectivity modeling workflow contributes to improved predictions, and assess the improvements of using a mechanistic over a simplistic model. We compile an extensive set of remote sensing data that accurately represent seasonality across the Okavango Delta ecosystem. These data are combined with dispersal GPS data from 30 wild dog coalitions in northern Botswana. We

then apply k-fold cross-validation for case-control studies to assess predictive efficacy under each configuration (Figure 1b) and model type. Lastly, we use iSSF-based simulations to estimate connectivity under different degrees of dynamism and both the simplistic and mechanistic models. To distinguish between different configurations (Figure 1), we assign codes that reflect the level of dynamism in each configuration (e.g., SSS = Static-SingleSeason-Static and DMD = Dynamic-MultiSeason-Dynamic). We hypothesize that habitat selection and movement behavior of dispersing wild dogs differs between seasons and that increasing seasonal dynamism improves the predictive efficacy (Figure 1c). We also anticipate that a mechanistic movement model yields better predictions of connectivity than a simplistic one.

2 Methods

We used the R programming language (R Core Team, 2023) for all data preparation and analyses. We performed spatial data manipulation using the `terra` (Hijmans et al., 2024) and `spatstat` (Baddeley et al., 2015) packages. We generated figures using the `ggplot2` (Wickham et al., 2024) and `ggpubr` (Kassambara, 2024) packages. To ensure reproducibility of all our analyses, we provide access to our R-scripts through an online repository upon publication of this article.

2.1 Study Area

The study area comprised the Okavango Delta ecosystem in northern Botswana (centered at 24°30'E and 20°42'S at an elevation of approx. 950 m) and extended to parts of Namibia and Zimbabwe, encompassing an extent of 160,000 km² (Figure 2). The Okavango Delta is a flood-pulse driven mosaic of patchy woodlands, permanent swamps, and seasonally flooded grasslands that lie within the otherwise dry and sandy Kalahari Basin (Wilson and Dincer, 1976; Ramberg et al., 2006; Mendelsohn et al., 2010). Precipitation across the study area varies considerably between seasons, ranging from 0 mm during the dry season (from 15 April to 15 October) to 140 mm during the wet season (from 15 October to 15 April), totaling to 600 mm across an average year (Figure 3a). Daily maximum above-ground temperature fluctuates between 7°C during the dry winter months and 38°C during the wet summer months (Figure 3b). The vegetation in the study area is mainly composed of mopane forest (*Colophospermum mopane*), mixed woodland acacia-dominated (*Acacia spp.*), and grassland. Substantial vegetation green-up (e.g., plant and shoots growth, leaf production, and grass greening) after the dry season starts with a delay of some weeks after the onset of the first rains in the wet season. The normalized difference vegetation index (NDVI)

therefore depicts a smoothed and lagged response to precipitation patterns across the study area (Figure 3c). The yearly flood-cycle of the Okavango Delta is predominantly driven by rainfalls in the Angolan highlands, where water is collected and channeled through the Cubango and Cuito rivers into the Okavango Delta (McCarthy et al., 2003; Gumbrecht et al., 2004; Mendelsohn et al., 2010). Because water only slowly descends from the catchment areas in Angola into the Delta’s tributaries, the flood is out of sync with local rainfalls and reaches its maximum extent between August and September, i.e. during peak dry season (Wolski et al., 2017, Figure 3d). While the extent of large-bodied rivers and floodplains is determined by precipitation in Angola, the emergence of smaller, ephemeral water-bodies (i.e., pans) is dictated by local precipitation during the wet season. 62% of the landscape in the study area form part of a protected area, such that human impact remains low and largely limited to settlements along the western part of the Okavango Delta and the city of Maun at the Delta’s southern tip (Figure 2). Landscapes outside protected areas in Zimbabwe, however, are more heavily influenced by humans, mainly through agricultural fields and human settlements.

2.2 GPS Data

Between 2015 and 2022, we collected GPS data on 30 dispersing wild dog coalitions (15 female coalitions, 15 male coalitions; Figure 2b). We programmed GPS satellite collars to record GPS locations at 01:00, 05:00, 09:00, 17:00, and 21:00. The eight-hour window between 09:00 and 17:00 can be considered comparable to a four-hour window during the rest of the day, as wild dogs generally rest between 11:00 and 15:00, leaving approximately four hours of activity (Hayward and Slotow, 2009). The collected GPS data were regularly transmitted to a base-station via Iridium satellite, thereby allowing dispersing individuals to be monitored, even if they ventured across national borders. In total, we obtained 5,940 locations during dispersal, with an average of 198 ± 239 locations per coalition. Occasionally, the acquisition of a GPS location failed (success rate = $93 \pm 8\%$), resulting in slight deviations from the aspired four-hourly schedule. Further details on the GPS collar fitting procedure and how we distinguished between dispersal and resident movements can be found in Cozzi et al. (2020) and Hofmann et al. (2021).

2.3 Covariates

We represented the physical landscape through which dispersers moved by a suite of spatial covariate layers known to influence wild dog movements during dispersal. We broadly

categorized these into descriptors of (1) landscape characteristics, (2) climatic conditions, and (3) anthropogenic features (Table 1; see Hofmann et al., 2021, 2023). Besides spatial covariates, we also prepared a series of covariates relating to (4) light availability (Table 1; see Cozzi et al. 2012). To appropriately render seasonality in each of the spatial covariates (1-3), we downloaded them at the highest spatial and temporal resolutions available. That is, we obtained for each spatial covariate a stack of raster layers that spanned the entire range of dates for which we also collected GPS data on dispersing wild dogs. Depending on the covariate, this resulted in differing spatial and temporal resolutions (Table 1). Additionally, for each spatial covariate, we also generated a static layer, representing “average” conditions across the entire duration of the study. For this, we flattened each covariate stack into a single layer, thus removing seasonality from the spatial data entirely. For continuous covariates, we achieved this by averaging conditions across all collected layers, whereas for categorical (binary) layers we identified areas that were covered by the respective category in at least 50% of all layers. Using the same aggregation techniques, we computed covariate stacks representative of a typical year. That is, instead of removing seasonality by flattening across the entire range of dates, we flattened stacks across years, thereby eliminating year-specific effects while retaining seasonal variation. These latter layers were later used in the dispersal simulation, where we needed spatial layers that depicted seasonal dynamism across a typical year. In summary, we prepared each covariate statically, dynamically for the entire range of dates considered, and dynamically for an average year.

2.3.1 Landscape Characteristics

We used data from the MODIS Vegetation Continuous Fields dataset (MOD44B V061; DiMiceli et al., 2022) to represent different vegetation types across the study area. The MOD44B dataset comprises three continuous layers, depicting the percentage cover of woodland, shrubs/grassland, and bareland, respectively. The three layers added up to 100%, so we dropped bareland from further analysis, thus preventing perfect multi-collinearity. The MOD44B product is updated on day 65 of each year, so we used the R-package `RGISTools` (Pérez-Goya et al., 2020) to download yearly updated layers, each at a resolution of 250m x 250m. We also obtained information on the NDVI through the MODIS MOD13Q1 dataset (Didan, 2015), which also has a resolution of 250m x 250m. This product is updated every 16 days, and we accessed the respective data through Google Earth Engine (Gorelick et al., 2017) using the R-package `rgee` (Aybar et al., 2024). To depict large, permanent waterbodies, we employed the Globeland30 dataset, from which we only retained the land-cover

class water, while setting all other categories to dryland. Similarly, we used the MERIT Hydro dataset to obtain information on permanent rivers (Yamazaki et al., 2019). To dynamically render large water-bodies, particularly the floodwaters of the Okavango Delta, we prepared weekly updated floodmaps using remote sensed MODIS MOD43A4 satellite images. The underlying floodmapping algorithm is described in detail by Wolski et al. (2017) and Hofmann et al. (2021) and is implemented in the `floodmapr` package (available on GitHub; <https://github.com/DavidDHofmann/floodmapr>). We combined the water (static), river (static), and flood layers (dynamic) into a single stack representing water bodies at a resolution of 500m x 500m (Table 1). Finally, we employed remote sensing to detect small, ephemeral water bodies (i.e., pans) using a custom random-forest classifier applied to Sentinel 2 satellite imagery (European Space Agency, 2018; details of the classifier in Appendix A1). Sentinel 2 has a resolution of 10m and is therefore particularly useful for obtaining information on small landscape features. Even though Sentinel 2 satellite imagery is updated every 5 days, cloud cover often prohibited the computation of a “pan-map”. Consequently, we settled for monthly updated composite images, which effectively alleviated problems due to cloud cover. In summary, we produced one stack of layers representing major water bodies, and another stack of layers representing ephemeral water bodies (pans). For both, we also computed corresponding distance-to layers, indicating the distance to major and ephemeral water bodies, respectively.

2.3.2 Climate Descriptors

We obtained hourly updated spatial layers on 2m above-ground temperature from the ERA5-Land dataset (Muñoz-Sabater et al., 2021) and hourly updated precipitation estimates from the Global Satellite Mapping of Precipitation dataset (Kubota et al., 2020). Both datasets were accessed and downloaded through Google Earth Engine (Gorelick et al., 2017) using the `rgee` package (Aybar et al., 2024) and had a resolution of 10km x 10km. To match the one-hourly temperature and precipitation estimates with the four-hourly GPS data on dispersing wild dogs, we computed average precipitation and temperature values over four-hourly periods that matched the GPS collection schedule.

2.3.3 Anthropogenic Features

We combined information on human density, agricultural activities, and roads into a single proxy, which we generically termed human influence. We sourced information on human density from Facebook’s high resolution human density dataset (Tiecke et al., 2017), which

we downloaded from the humdata website (www.data.humdata.org). We obtained information on the presence of agricultural fields from the Globeland30 (Chen et al., 2015) and Cropland (Xiong et al., 2017) datasets. We downloaded shapefiles comprising main tar roads from OpenStreetMaps (OpenStreetMap contributors, 2017). Finally, we merged all anthropogenic features into a single layer that had a resolution of 250m x 250m (details are provided by Hofmann et al., 2021) .

2.3.4 Light Availability

We computed light availability using the `suncalc` and `moonlit` R-packages (Thieurmél and Elmarhraoui, 2022; Śmielak, 2023) for the central coordinates of our study area at a 5-minute temporal resolution. The set of light statistics comprised a binary variable separating day and night (i.e., $\text{sun} < -18^\circ$ below the horizon) and a continuous estimate of moonlight illumination, relative to the maximum moon illumination. Based on those covariates, we generated a binary covariate separating bright from dark conditions. Bright conditions encompassed all daytime hours and those nighttime periods during which the moon was present in the sky and illuminated by approximately one-fourth; conversely, dark conditions included nighttime periods during which the moon was absent from the sky or present and only minimally illuminated (further details in Appendices A2 and A3).

2.4 Step-Selection Models

We modeled dispersers' movement kernels and habitat selection using integrated step-selection functions (iSSFs, Fortin et al., 2005; Avgar et al., 2016), following the procedures described by Muff et al. (2020) and Fieberg et al. (2021). For this, we identified bursts of subsequent GPS locations where the duration between two locations did not exceed 4 hours (± 15 minutes) or 8 hours (± 30 minutes, for data recorded between 09:00 and 17:00). Within each burst, we converted locations into steps, where a step represented the straight line segment between two consecutive locations (Turchin, 1998). For each step, we computed the associated step length (sl, in meters) and relative turning angle (ta, in radians). During this pre-processing, four coalitions dropped due to temporally irregular data, leaving a total of 26 dispersing coalitions (12 female coalitions, 14 male coalitions) for further analyses. The final dataset comprised 5,310 steps (204 ± 225 per coalition), which we further categorized into steps collected during the wet season (15 October to 15 April) and steps collected during the dry season (15 April to 15 October), yielding 3,124 steps (59%) during the dry season and 2,186 steps (41%) during the wet season.

We paired each *observed* step with a set of 100 *random* steps, generated by sampling turning angles from a uniform distribution $U(-\pi, +\pi)$ and step lengths from a gamma distribution fitted to observed steps (scale $\theta = 6341.96$ and shape $k = 0.39$). Together, an observed step and its associated random steps formed a stratum that received a unique identifier. Along each step, we extracted covariate values from the underlying spatial layers, and determined light conditions (Table 1). We opted for covariate extraction *along* steps rather than *at their endpoints*, as we believed that environmental conditions along steps were relevant in determining wild dog movement during dispersal (sensu Hofmann et al., 2023). For continuous covariates, we computed average values along each step, for categorical covariates the percentage cover of each category along the step. To model a decreasing marginal impact of “distance-to” variables, we included their square-root as predictors in the final models. To facilitate model convergence, we normalized extracted values to a range between zero and one.

To estimate movement and habitat parameters of interest, we applied the maximum likelihood procedure proposed by Muff et al. (2020), using the `glmmTMB` package (Brooks et al., 2017). To examine whether the effect of accounting for seasonality was modulated by the model structure, we employed two models informed by previous studies on wild dog dispersal (Hofmann et al., 2021, 2023). In the *simplistic model*, we exclusively included covariates relating to spatial features, alongside the movement descriptors (sl , $\log(sl)$, $\cos(ta)$) that are mandatory to fit an iSSF (Avgar et al., 2017; Fieberg et al., 2021). This formula is akin to the models that are used with resistance-based approaches, which prohibit a mechanistic understanding of movement. In the *mechanistic model*, by contrast, we considered additional interactions between step descriptors, landscape characteristics, climate descriptors and light availability, rendering that wild dog movement during dispersal may be affected by such interactions. In both models, we also included a stratum-specific intercept with a large fixed variance (10^6 , Muff et al., 2020), and used dispersing coalition ID to model random slopes for all main effects. To ensure comparability across the six configurations of seasonal dynamism (Figure 1), we applied the same model structures but varied the data used for model fitting (Figure 1a). In configuration SSS, we used static environmental data and pooled movement data across seasons. In configuration SMS, we used static environmental data, but split the movement data by season. In configurations DSS and DSD, we used dynamic covariates but pooled movement data. In configurations DMS and DMD, we used dynamic covariates and split movement data by season. For each configuration, we fit both a simplistic and mechanistic model. To quantify how many random steps were necessary

before model estimates stabilized (sensu Fieberg et al., 2021), we fitted each model with 5, 10, 25, 50, 75, and 100 random steps.

2.5 Validation

We compared the predictive efficacy of each of the six configurations presented in Figure 1b using k-fold cross-validation for case-control studies (Fortin et al., 2009). For this, we split movement data into training and validation sets using an 80:20 ratio and fit the iSSF models described above. We then used the obtained β -estimates to predict the probability of each random and observed step in the validation set for being chosen. Depending on the configuration, we predicted step-probabilities using static or dynamic covariates. Within each stratum, we assigned ranks 1-101 to each step based on predicted probabilities and recorded the number of times the observed step was assigned each rank. Finally, we computed Spearman’s rank correlation between ranks and associated frequencies $r_{s,realized}$. The better a model’s predictive ability, the more negative $r_{s,realized}$ should be (i.e., the less often the observed step should be assigned a low rank). For reference, we also computed Spearman’s rank correlation for randomized preferences ($r_{s,random}$), which we achieved by removing the observed step from each stratum and identifying the rank of a randomly chosen step. We replicated this validation procedure 100 times and employed ANOVA to compare the predictive efficacy of different configurations.

2.6 Simulations

To assess differences in predicted connectivity upon increasing seasonal dynamism and depending on the model type, we ran dispersal simulations under the two most distinct configurations, i.e. the fully static (SSS, Figure 1) and fully dynamic (DMD, Figure 1) configurations, once using the simplistic model, once using the dynamic model. We simulated individuals starting from three distinct regions known to host viable wild dog populations (Figure 2). The definition of these areas was somewhat arbitrary, albeit we deliberately selected areas in the west and east of the Okavango Delta to examine potential influences of flooding on connectivity (sensu Hofmann et al., 2024). We also selected a more independent location (Nxai Pan; Figure 2) that presumably remained unaffected by the Okavango Delta’s flood extent. To initiate simulated dispersal trajectories, we randomly placed 1,000 start points within each of these source area, with start times that were equally distributed across the year. To simulate dispersal and obtain connectivity maps under both configurations, we applied the simulation algorithm for iSSFs described by Signer et al. (2017) and

employed by Hofmann et al. (2023). A similar algorithm has recently been added to the **amt** R-package (Signer et al., 2024). We applied the simulation algorithm as follows. Originating from the start point, we generated a set of 25 random steps by sampling step lengths from the fitted gamma distribution and turning angles from a uniform distribution. Along each random step, we extracted spatial covariates, computed relevant movement metrics (sl, $\log(\text{sl})$, $\cos(\text{ta})$), and assigned light conditions. We then employed the fitted iSSF model to predict the probability of each step for being chosen, thus generating a “redistribution kernel” around the current location of the simulated individual (Signer et al., 2017, 2024). Based on this kernel, we sampled one of the steps and updated the simulated individual’s position and time. We repeated the procedure until a total of 2,000 steps were realized (~ 400 dispersal days).

Depending on the configuration, we employed different spatial covariates and preferences during the simulation. For the SSS configuration, we omitted any seasonality, and predicted connectivity using the static set of covariates and fixed preferences (i.e., β -estimates). For the DMD configuration, on the other hand, we updated covariates and preferences dynamically as the simulation progressed. That is, after every simulated step, we generated a new covariate stack comprising spatial layers that best represented environmental conditions at that particular point in time (Figure 4). For the purpose of this comparison, we employed the set of covariates representing an average year. Similarly, we predicted the redistribution kernel using β -estimates from the appropriate season. Finally, we obtained and compared heatmaps under both configurations.

3 Results

Due to convergence issues, we removed the covariates NDVI, precipitation, and distance to pans from all analyses. Some minor convergence issues remained and our validation procedure failed in 59 out of 1,800 validation attempts. Since failures occurred randomly across different configurations, we deemed their exclusion of no concern.

3.1 Step-Selection Models

Patterns of habitat selection and movement behavior were qualitatively similar, irrespective of whether models were fit using static or dynamic covariates (Figure 5, Tables S3-S6) and irrespective of the number of random steps considered (Figure S9). The most notable quantitative difference when using static vs. dynamic covariates was that models fitted using dynamic covariates resulted in narrower confidence intervals (Figure 5). Furthermore, effect

sizes (i.e., β -estimates) of models fit using static covariates were more pronounced than effect sizes from models that were fit using dynamic covariates (Figure 5). Differences were most marked for $\beta_{DistanceToWater^{0.5}}$, which was estimated ≈ -0.8 using static covariates, but ≈ -0.3 when using dynamic covariates.

Differences in β -estimates between seasons were moderate and most pronounced when models were fit using static covariates. For instance, when models were fit using static covariates, β_{Water} was ≈ -0.9 during the dry season but ≈ 0.2 during the wet season. When dynamic covariates were used, by contrast, β_{Water} was ≈ -0.6 and ≈ -0.3 , respectively. $\beta_{sl:Dark}$ was also markedly lower during the wet season (≈ -0.9), compared to the dry season (≈ -0.1) but this was independent of the type of covariates that were used. Plots that aid with the interpretation of the models are provided in Figure S10.

3.2 Validation

Spearman’s rank correlation coefficients (r) obtained from the validation procedure revealed that predictions from the mechanistic model performed markedly better than those from the simplistic formula ($\bar{r}_{simplistic} = -0.5$, $\bar{r}_{mechanistic} = -0.9$, Figure 6). Irrespective of the employed model, Spearman’s rank correlation differed significantly depending on the amount of dynamism considered (simplistic: $F(5, 593) = 26.45$, $p < 0.001$, mechanistic: $F(5, 594) = 7.14$, $p < 0.001$), albeit with moderate effect sizes. Using the simplistic formula, moving from a fully static (SSS) to a fully dynamic (DMD) configuration decreased Spearman’s rank correlation (i.e., increased the predictive performance) by 0.15 from -0.41 to -0.56. With the mechanistic model, conversely, moving from a fully static (SSS) to a fully dynamic (DMD) configuration entailed an increase in r (i.e., a decrease in the predictive performance) by 0.01 from -0.89 to -0.88. Our hypothesis that accounting for seasonal dynamism results in better predictive performance only held when using the simplistic model (Figure 6a), but not when employing the mechanistic model (Figure 6b).

3.3 Simulations

Simulations under the most static (SSS) and dynamic (DMD) configurations revealed different connectivity patterns (Figure 7). In the static configuration, simulated trajectories were more concentrated, suggesting frequent movement across a few key habitats. In the dynamic configuration, by contrast, simulated trajectories were more homogeneously distributed, suggesting that movement was less restricted. When qualitatively comparing the heatmaps generated under the simplistic and mechanistic models, differences in predicted

connectivity are imperceptible.

4 Discussion

4.1 Brief Summary

We introduced a conceptual framework to highlight that seasonality can enter a connectivity analysis at three stages; (1) when extracting spatial covariates for fitting the selection model, (2) when fitting the selection model, and (3) when predicting from the fitted model. Through combination, this yields six configurations that differ in their degree of dynamism and, arguably, realism. We fit the models and ran the dispersal simulations associated with each configuration using movement data on dispersing wild dogs, and employed a rigorous validation procedure to investigate potential gains in predictive performance that can be reaped by incorporating different levels of seasonal dynamism. Estimates from the fitted models showed that including seasonality only marginally affected inferred patterns of habitat selection and movement behavior. Similarly, the validation procedure attested only moderate improvements in predictive performance upon increasing the degree of seasonal dynamism. Crucially, these benefits were limited to an overly simplistic model and vanished upon fitting a mechanistic one. In fact, the mechanistic model resulted in such high predictive efficacy that increasing seasonal dynamism only marginally improved predictions. Despite this, we found that simulations under the most static and most dynamic configurations resulted in contrasting connectivity patterns. Under the most static configuration, landscape connectivity was heterogeneous and clumped around a few hot spots. Under the most dynamic configuration, where species preferences and environmental conditions changed as the simulation proceeded, connectivity was more homogeneously distributed.

4.2 Moderate Improvements upon Increasing Dynamism

Our validation procedure revealed only moderate improvements in our ability to predict dispersal movements when increasing the degree of seasonal dynamism. Given the system's extreme seasonal variability, this was somewhat surprising. We suspect that the absence of a more pronounced improvement can be traced back to multiple factors. Firstly, we focused our analysis on dispersing individuals, which cover large distances in search of potential mates and a suitable territory (McNutt, 1996; Cozzi et al., 2020). In our case, the average 4-hourly step length was 2.5 km, implying that dispersers cross and sample a large variety of habitats within short time. The spatial scale of the dispersal process may thus misalign

with the spatial scale at which seasonality affects connectivity. A further explanation could emerge from the fact that dispersers prioritize finding unoccupied territories and areas with low competition, rather than selecting specific habitat types (e.g., Creel and Creel, 1996; Creel, 2001). Dispersers’ habitat selection may therefore be more strongly influenced by territorial considerations (*sensu* Cozzi et al., 2018) and only little affected by seasonally changing landscape characteristics. This would also explain the relatively weak selection or avoidance of environmental characteristics exhibited in Figure 5. Finally, the African wild dog is a generalist species that occupies a broad variety of habitats (Woodroffe, 2011). A certain tolerance towards changing environmental conditions can therefore be expected and may explain why seasonal differences were faint. This holds particularly true for dispersers, which are usually more tolerant towards unfavorable habitat conditions as they spend little time within the same area (Elliot et al., 2014; O’Neill et al., 2020). Even though incorporating seasonality provided little benefit in terms of predictive performance in our system, adopting our conceptual framework facilitates greater clarity and rigor in connectivity studies. By explicitly defining where and how seasonality enters the modeling pipeline, our framework encourages researchers to be more specific about the type of seasonal variation their study accounts for, reducing ambiguity in how seasonal effects are represented. Additionally, it prompts a more deliberate consideration of the pathways through which seasonality can influence connectivity, helping to identify whether its effects primarily stem from environmental variability, shifts in species behavior, or both. This approach not only enhances transparency and comparability across studies but also fosters a more mechanistic understanding of movement, ultimately improving the ecological relevance of connectivity analyses.

4.3 Substantial Improvements upon Fitting a Mechanistic Model

Compared to the improvements achieved by increasing seasonal dynamism, moving from the simplistic to the mechanistic model more significantly benefited predictions. Through the mechanistic model, we could represent several important aspects of African wild dog biology, such as reduced movement during dark nights (Cozzi et al., 2013) and times of high ambient temperature (Rabaiotti et al., 2021). We could also model that dispersers’ movement behavior depended on habitat conditions, as evidenced by shorter steps in areas covered by water (Hofmann et al., 2023). The ability of encapsulating and replicating dispersal movements in such detailed and mechanistic manner is unique to iSSF-based simulations and cannot be achieved using resistance-based models. This is also a testimony to step-selection

functions as a powerful means of conducting both inference and prediction of movement and connectivity (Hofmann et al., 2023; Forrest et al., 2024). While many of the additionally encapsulated processes are not directly linked to landscape connectivity, incorporating them helps control for key behavioral and environmental influences that might otherwise obscure critical movement patterns. In our case, this versatility elevated the predictive power of our dispersal model to a level where the added benefit of considering seasonal dynamism was negligible. However, developing mechanistic models requires a deep understanding of the study system, which can only be gained through direct field exposure. Recent critiques have pointed out a growing tendency within the scientific community to undervalue fieldwork (Rafiq et al., 2024), yet our findings reinforce its importance for building ecologically and biologically meaningful models.

4.4 Dynamic Connectivity

Comparisons of connectivity under the most static (SSS) and dynamic (DMD) configurations, revealed that connectivity was concentrated on a few major dispersal hotspots under the most static configuration, but uniformly distributed across the entire landscape under the most dynamic configuration. This corroborates previous research showing that static connectivity analyses tend to underestimate connectivity, as seasonal stepping stones or corridors are missed (Martensen et al., 2017). Via seasonally available dispersal habitats, dispersers periodically gain access into otherwise inaccessible areas. In our case, such stepping stones arose in two ways; via changes in the landscape, which caused a rearrangement of preferred habitats, and via changing preferences of the dispersing individuals. Jointly, seasonal landscapes and changing preferences resulted in a more balanced mosaic of connectivity. Note, however, that the fact that connectivity was more evenly distributed across the year does not exclude the possibility of certain areas experiencing lower connectivity seasonally. As already shown by Osipova et al. (2019), a static representation may result in both an over- and under-estimation of short-term connectivity, depending on the area and season. For conservation planning, this implies a need to protect landscapes more broadly, as areas that provide little connectivity during some season may be critical during others. A dynamic take at connectivity therefore improves our understanding of ecological processes in dynamic landscapes and may help to identify otherwise overlooked dispersal hotspots (Zeller et al., 2020a).

4.5 Validation Procedure

Our results showed only marginal improvements in predictive efficacy with increased seasonal dynamism, yet revealed vastly different connectivity patterns. While this may seem contradictory, it likely arises from our validation approach, which was constrained to the step level due to a fundamental limitation in validating seasonal connectivity from an iSSF. In dynamic landscapes with shifting preferences, connectivity varies both spatially and temporally, requiring validation at each timestamp in the dataset. Since our landscape changed every four hours (i.e., with each step), a new connectivity map would need to be generated around each validation step, effectively reducing validation to the step level. Given this constraint, we deemed the application of k-fold cross-validation using Spearman’s rank correlation at the step level appropriate. Note, however, that Spearman’s rank correlation is influenced by the number of steps per stratum, making it unsuitable for comparing predictive efficacy across studies (Figure S11). For static connectivity analyses, alternative validation techniques exist, which typically compare connectivity at observed vs. random locations (McClure et al., 2016; Zeller et al., 2018; Brennan et al., 2020; Finerty et al., 2023). Developing similar validation approaches beyond the step level for dynamic connectivity analyses remains a challenge for future research.

4.6 The Costs of Incorporating Dynamism

Increasing seasonal dynamism to model dynamic connectivity comes at significant costs, both in terms of data requirements and computational challenges. To represent seasonal changes across the landscape, spatial layers need to be downloaded and processed at an appropriate seasonal interval. This is time-consuming and implies a substantial increase in the data-volume to be handled. Seasonal products are less commonly available, potentially necessitating custom remote-sensing analyses (e.g., Appendix A1). Moreover, seasonal layers are often plagued by noise and missing data, especially during seasons with extensive cloud cover. Extracting covariates from seasonal layers poses a further challenge, as covariate data can no longer be extracted from single layers, but need to be extracted from those layers that best represent environmental conditions at the time of the observed movement. This applies to the extraction of covariates for both model fitting (Figure 1, stage 1) and prediction (Figure 1, stage 3). Finally, splitting species data by season to model seasonal preferences reduces the amount of data remaining per season, potentially causing convergence issues. In our case, the moderate improvement in predictive ability did not warrant these additional efforts and complications. The trade-offs between increased realism and

practical applicability must be carefully considered, as greater complexity does not always lead to better insights and may even obscure key ecological patterns (Puy and Saltelli, 2022).

4.7 Conclusion

In conclusion, we provided a conceptual framework to integrate seasonality into connectivity analyses and investigated the role of seasonal dynamism in modeling dispersal and connectivity. While increasing the level of dynamism led to only moderate improvements in predictive performance, it had a substantial impact on inferred connectivity patterns, resulting in more evenly distributed connectivity when seasonality was considered. This suggests that seasonal fluctuations can shape connectivity in ways that may not be immediately evident from overall predictive accuracy alone. However, the added complexity of incorporating seasonal landscapes and species preferences must be carefully weighed against its practical benefits. Beyond the effects of seasonal dynamism, our findings highlight that model structure plays a crucial role in shaping movement predictions. In our case, a biologically informed mechanistic model, which explicitly accounted for species-specific movement constraints, outperformed a simplistic model, akin of those typically used with resistance-based approaches. In fact, a mechanistic model obliterated any improvement of accounting for seasonality by elevating predictive efficacy to a very high level. While this result may be specific to our study system, it underscores the broader importance of selecting an appropriate movement model when studying connectivity and it attributes a large potential to iSSF-based simulations for studying connectivity.

5 Authors' Contributions

D.D.H. and G.C. conceived the study and designed methodology; D.D.H., G.C., D.M.B, and J.W.M. collected the data; D.D.H. analysed the data; G.C. assisted with modeling; D.D.H. and G.C. wrote the first draft of the manuscript and all authors contributed to the drafts at several stages and gave final approval for publication.

6 Data Availability

Access to R-scripts to replicate our analyses will be provided through an online repository at the time of publication.

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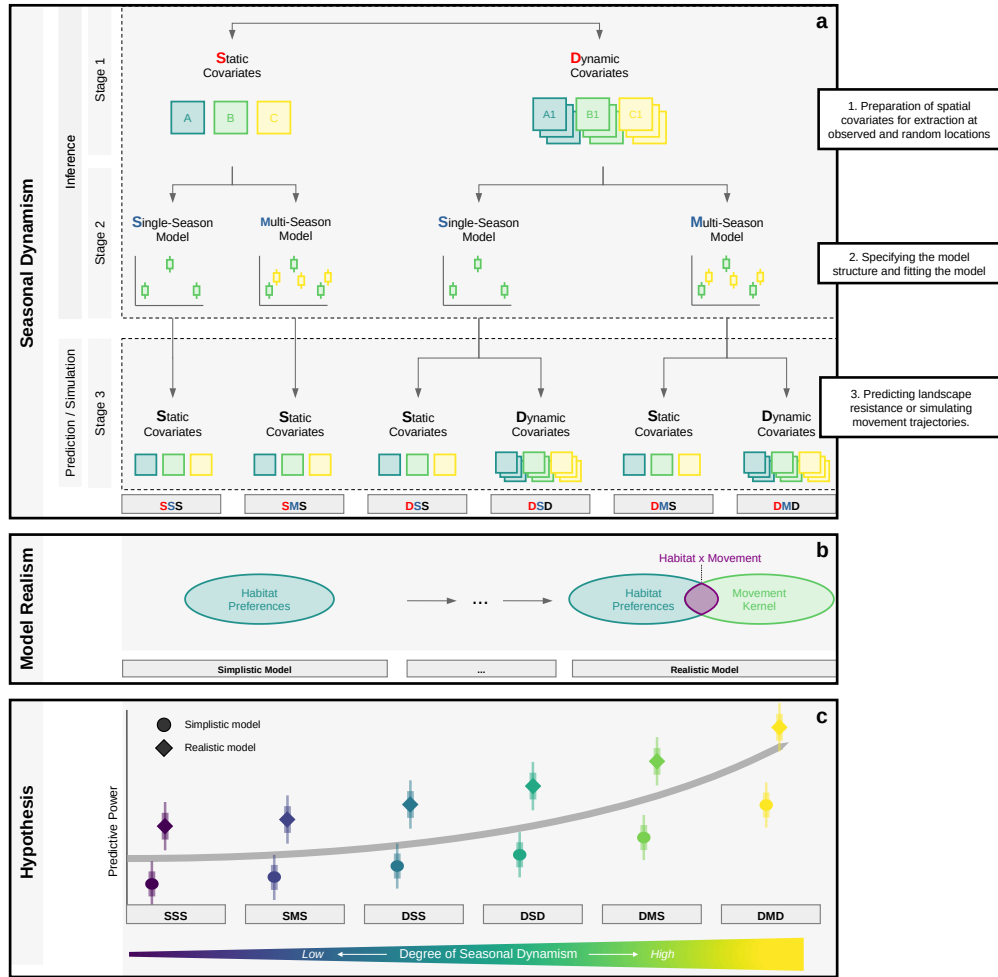


Figure 1: Overview of the different stages at which seasonality can be incorporated in studies of dispersal and landscape connectivity. (a) During model fitting, one needs to decide whether to represent the environment by a static set of covariate layers, thus ignoring seasonality, or to obtain a dynamic set covariate layers that allow capturing how the landscape changes over time. One also needs to decide whether to parameterize a single-season model, assuming fixed species preferences across the year, or to engage in a multi-season model that accounts for seasonal differences. (b) When utilizing the fitted model to predict connectivity, one can either assume a static set of environmental covariates or again attempt a seasonal take that renders how connectivity differs depending on the season. (c) Depending on these decisions, six different combinations that differ in terms of the degree of seasonality they encapsulate emerge. Our hypothesis was that increasing the degree of dynamism when predicting dispersal and connectivity would lead to improved predictions.

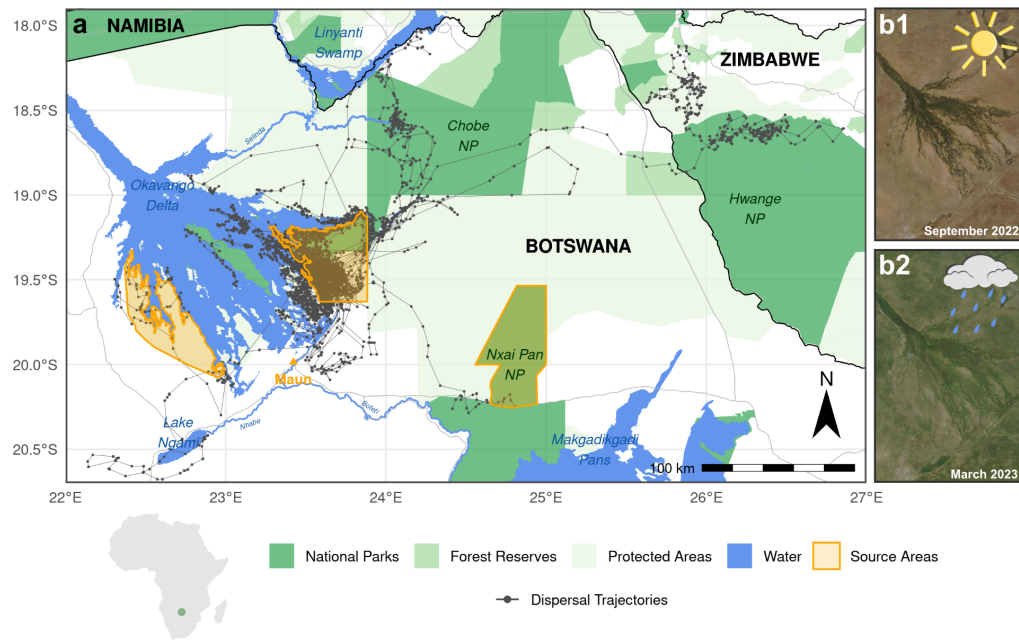


Figure 2: (a) Study area from which GPS data on dispersing African wild dogs were collected. Dispersal trajectories are displayed in dark gray. The study area encompassed parts of the Okavango Delta in northern Botswana, a highly dynamic, flood-pulse-driven ecosystem. The entire study area undergoes substantial seasonal changes, as can be seen from two satellite images taken during peak dry season (b1) and peak rainy season (b2). Notably, the flood of the Okavango Delta reaches its maximum extent during peak dry season between August and September.

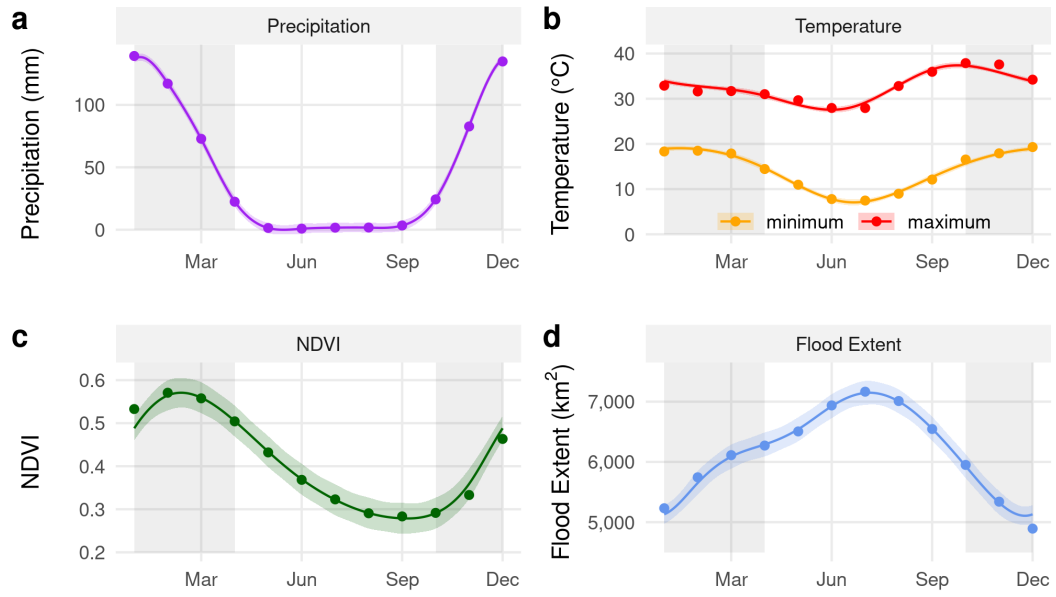


Figure 3: Illustration of how some of the covariates considered in this study vary across seasons. The wet season spans mid-October to mid-April (shaded in gray), the remainder is considered as dry season. Data for the graphs were obtained from (a) JAXA GSMP, (b) ERA5, (c) MODIS MOD13Q1, and (d) remote sensed MOD43A4 satellite images. Smoothing curves were fitted using GAMs as implemented in the *mgcv* R-package (Wood, 2011).

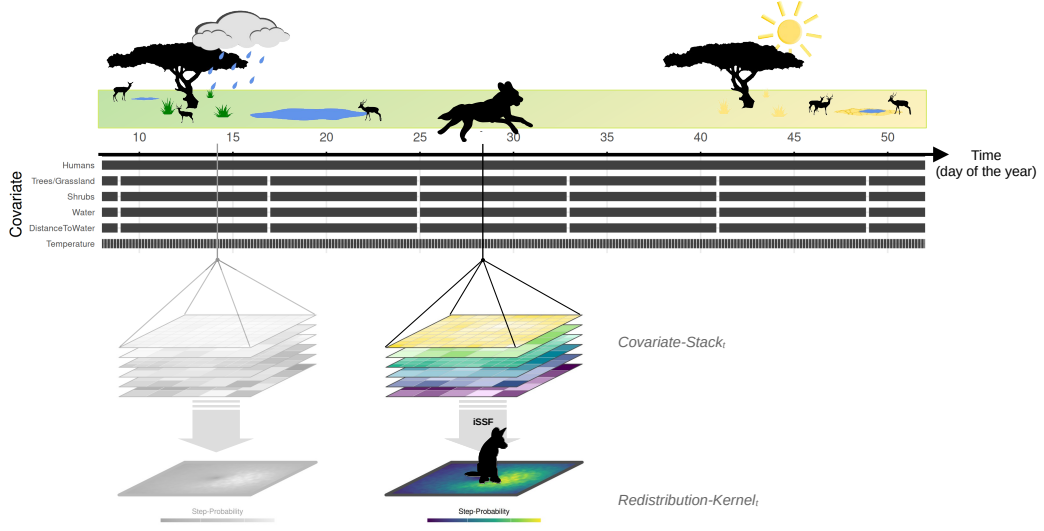


Figure 4: Schematic illustration of our dispersal simulation with dynamic covariates. As the simulation proceeded, the underlying spatial covariates (symbolized by the stack of layers) were updated. Depending on the covariate, the update frequency varied from a few hours (e.g., temperature) to multiple months (e.g., shrub cover). Each gray block represents a single layer and the duration for which it was “active”. Originating from the current position of the simulated animal, a redistribution kernel was derived, indicating potential steps and their predicted probability. We generated redistribution kernels by proposing a set of 25 random steps and applying the parametrized step-selection model to predict the probability of each step for being chosen. Based on this kernel, one location was randomly sampled and the animal’s position and time were updated. This procedure was then repeated until 2,000 steps (~ 400 dispersal days) were simulated.

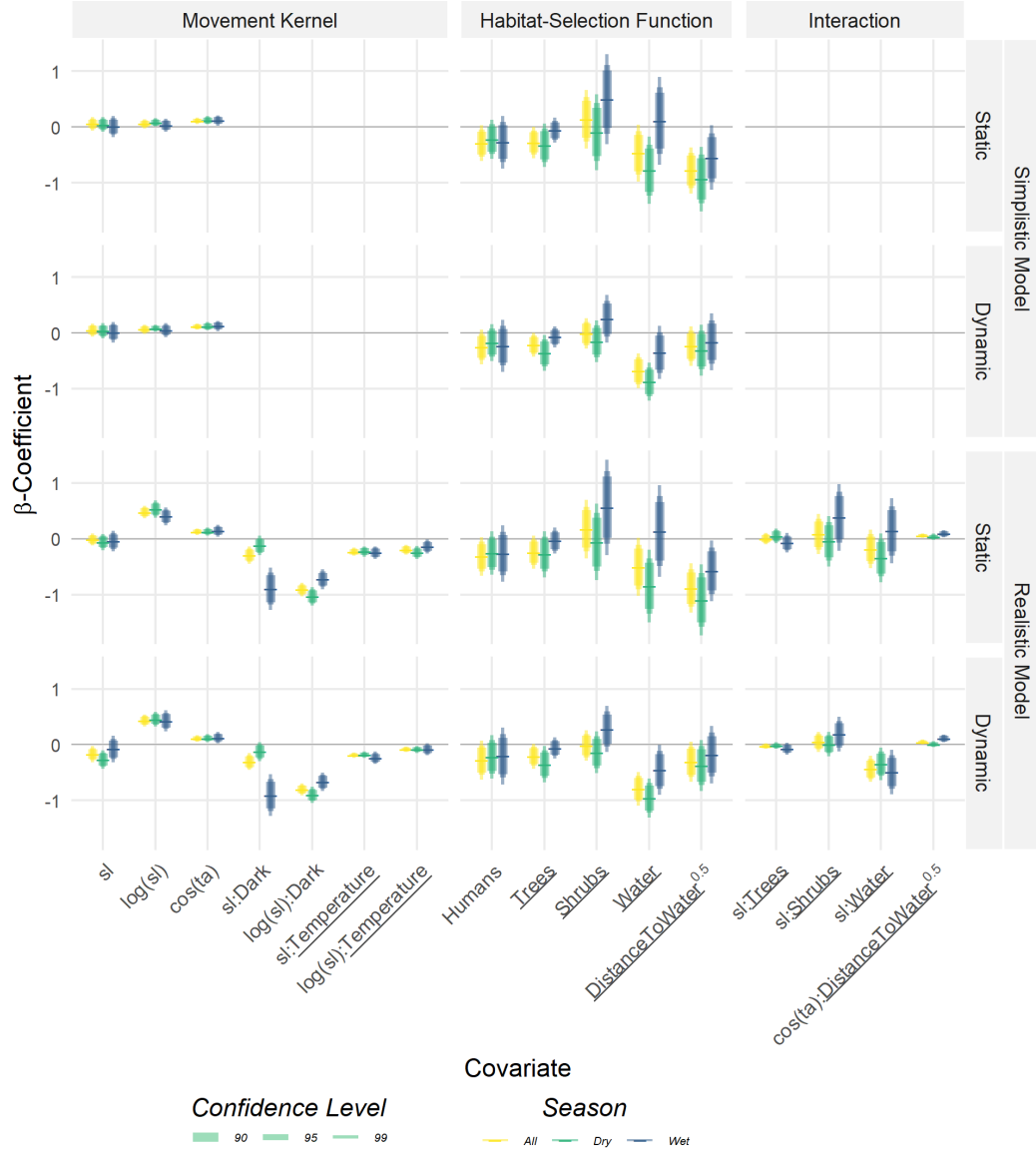


Figure 5: β -estimates from the integrated step-selection models, grouped by movement kernel, habitat-selection function, and their interaction. We either fit models using a *simplistic model* (without interactions) or a *mechanistic model* that comprised several interaction terms that rendered how movement behavior depended on environmental conditions. Furthermore, we distinguished between models fit using *static* and *dynamic* covariates. However, only the underlined covariates differed between the static and dynamic configurations, as covariates were either represented as a single layer (static), or a stack of layers (dynamic). Furthermore, data were either pooled across seasons (yellow bars) or split into dry (green bars) and wet (blue bars) season. Results are shown for replicates that used 25 random steps, as all associated models converged. The number of random steps did not appear to influence model estimates (Figure S9).

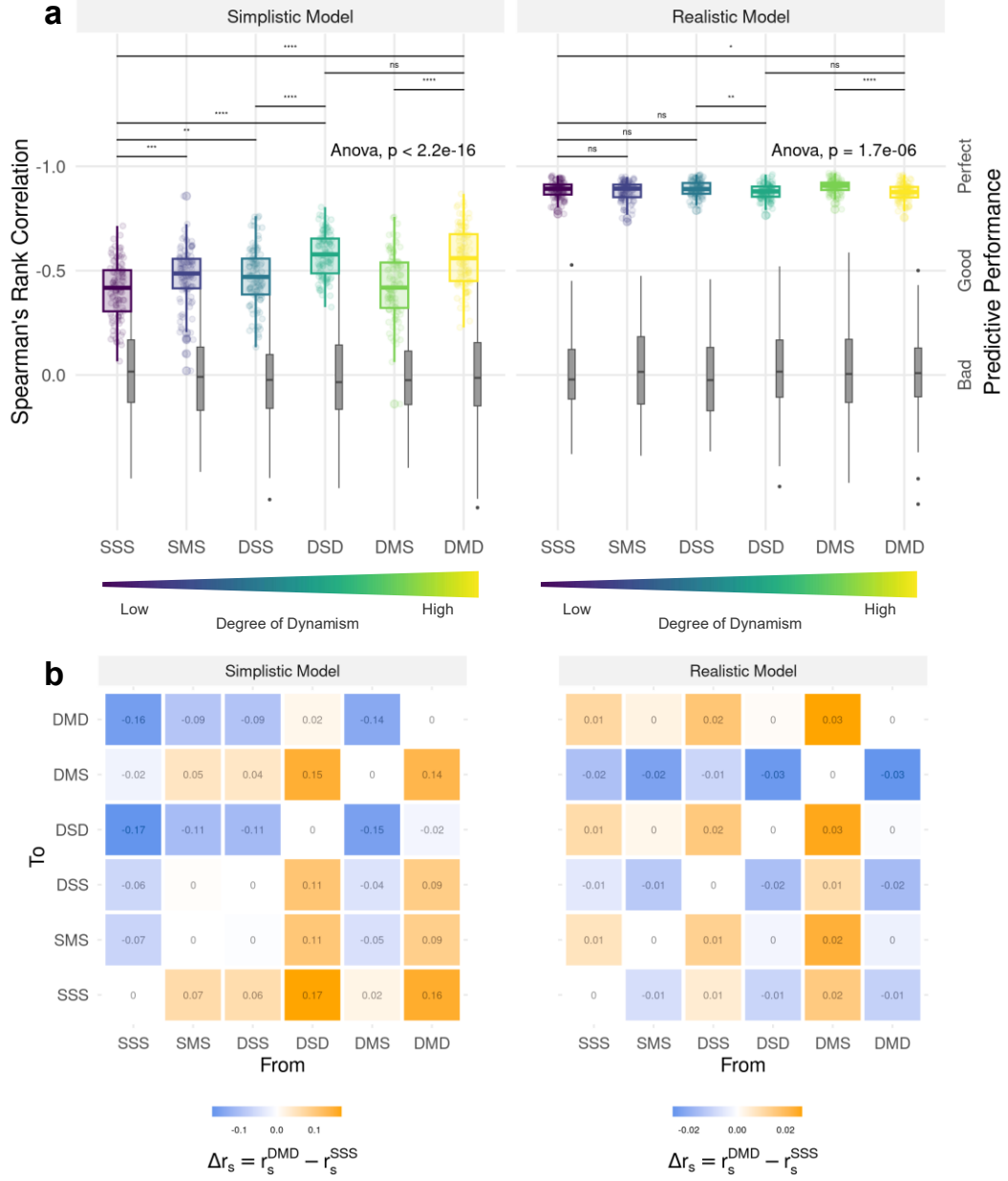


Figure 6: (a) Spearman's rank correlation across different configurations of dynamism that range from entirely static (SSS) to fully dynamic (DMD). The more negative Spearman's rank correlation, the better is the predictive performance under the respective configuration. Correlations were computed for 100 replicates. Note that the y-axis is inverted to match our expectation of increasing performance as dynamism increases. The boxplots in light gray represent Spearman's rank correlation coefficients under a null model with randomized preferences and serve for comparison. (b) Difference in Spearman's rank correlation when moving from one configuration to another. Values < 0 (blue) indicate an increase in predictive performance, whereas values > 0 (orange) indicate a decrease in predictive performance.

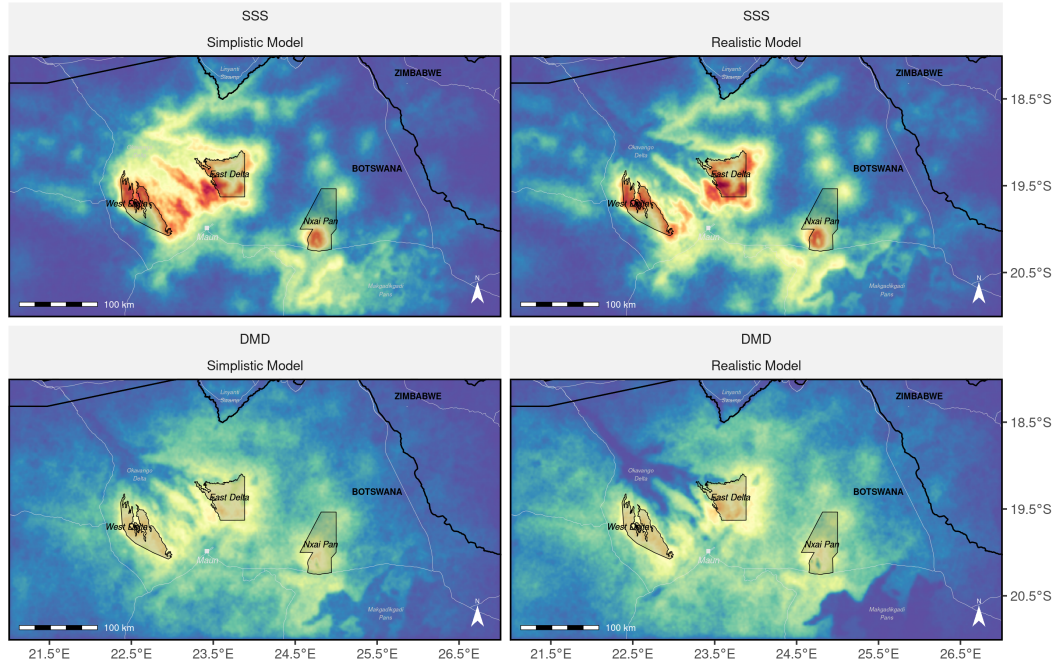


Figure 7: Heatmaps (i.e., number of traversing trajectories per pixel) derived under the most static (SSS) and dynamic (DMD) configurations both when simulating from the simplistic and mechanistic models.

Table 1: Covariates that were used in this study and information on their temporal and spatial resolutions, as well as on the avenue through which the respective data were accessed or downloaded. Download methods printed in **typewriter** font indicate the use of an **R**-package that is available through CRAN or GitHub.

Category	Variable	Type	Temporal Resolution	Spatial Resolution	Source
(1) Landscape Characteristics					
Landscape Characteristics	Trees	Continuous	1 Year	250 m	MODIS MOD13A2
Landscape Characteristics	Shrubs / grassland	Continuous	1 Year	250 m	MODIS MOD13A2
Landscape Characteristics	NDVI	Continuous	16 Days	250 m	MODIS MOD13A2
Landscape Characteristics	Rivers	Binary	Static	90 m	MERIT Hydro
Landscape Characteristics	Permanent water	Binary	Static	30 m	Globeland30
Landscape Characteristics	Floodwater	Binary	8 Days	500 m	MOD34A4
Landscape Characteristics	Distance to water	Continuous	8 Days	500 m	MOD43A4
Landscape Characteristics	Pans	Continuous	5/10 Days	10 m	Sentinel 2
Landscape Characteristics	Distance to pans	Continuous	5/10 Days	10 m	Sentinel 2
(2) Climate Descriptors					
Climate	Temperature	Continuous	4 Hours	10 km	ERA5
Climate	Precipitation	Continuous	4 Hours	10 km	JAXA GSMaP
(3) Anthropogenic Features					
Anthropogenic	Human density	Continuous	Static	30 m	Facebook
Anthropogenic	Agriculture	Binary	Static	30 m	Globeland30 / Cr
Anthropogenic	Roads	Binary	Static	vectorized	Open Street M
(4) Light Intensity					
Light Availability	Night	Binary	4 Hours	-	-
Light Availability	Moon illumination	Continuous	4 Hours	-	-

Note: The covariates in gray were combined into single proxies for water, human influence, and light availability. Detailed procedures are provided in Hofmann et al. (2021).